12TH BIOLOGY · CHAPTER 22 · FULL CHAPTER NOTES

Biostatistics

Topic-wise short and long questions with answers from the complete chapter

22.1 BIOSTATISTICS

Q.1 [Short] Define biostatistics and state its use.

Biostatistics is a scientific discipline that applies statistical methods and techniques to biological, medical and health-related data. It involves the collection, presentation, analysis and interpretation of data to draw meaningful conclusions. It allows researchers to make informed decisions about a whole population from sample data, because testing an entire population may be impractical or costly. For example, by studying a representative subset of workers, the prevalence of a health condition among all workers can be estimated.

Q.2 [Short] Why are samples studied instead of entire populations?

Testing an entire population may be impractical or costly. Biostatistical analysis lets researchers study a representative sample and make informed decisions about the whole population from it.

Q.3 [Long] Describe the components of biostatistics.

1. **Study design and experimental planning:** forming the research question and objectives; selecting the population, defining variables, randomization, controlling confounding factors and calculating sample size. Proper design gives valid, reliable and unbiased results.
2. **Data collection:** *primary data* come directly from experiments, surveys, interviews, observations and clinical trials; *secondary data* come from existing published research, medical records, government reports and databases.
3. **Data management and processing:** organizing the data systematically before analysis: cleaning, coding and storage, with confidentiality and integrity.
4. **Data analysis:** applying statistical techniques. *Descriptive statistics* describe the data (mean, median, mode, standard deviation). *Inferential statistics* (hypothesis testing, regression analysis, modelling) draw conclusions about populations from sample data.
5. **Interpretation of results:** translating statistical findings into meaningful biological or health conclusions, keeping in view the context, limitations and assumptions.
6. **Presentation and communication:** communicating results with charts, graphs and clear language; transparent reporting gives reproducibility and trust.

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Q.4 [Short] Differentiate between primary and secondary data collection.

Primary data collection	Secondary data collection
Data are collected directly by the researcher.	Existing data collected by others are used.
Sources: experiments, surveys, interviews, observations, clinical trials.	Sources: published research, medical records, government reports, databases.

Q.5 [Short] Differentiate between descriptive and inferential statistics.

Descriptive statistics describe the characteristics of the data, e.g., mean, median, mode and standard deviation. **Inferential statistics**, such as hypothesis testing, regression analysis and modelling, allow conclusions about populations from sample data.

Q.6 [Short] What is included in data management and processing?

Organizing the collected data systematically and ensuring accuracy before analysis. It includes data cleaning, coding and storage, while maintaining confidentiality and integrity.

Q.7 [Long] Describe the applications of biostatistics.

Application	Use of biostatistics
Clinical trials and medical research	Designing studies and analysing data; evaluating the safety and effectiveness of new drugs, treatments and devices
Epidemiology	Studying the distribution and determinants of diseases; how diseases spread; risk factors; effect of preventive measures
Genetics and genomics	Relating genes to diseases; inheritance patterns; genetic variation; personalized medicine
Public health policy	Analysing population health data; identifying vulnerable groups and health disparities
Environmental health	Assessing how pollution and toxins affect human health; guiding regulations
Bioinformatics	Analysing vast genomic and proteomic datasets for drug discovery and diagnosis
Healthcare quality control	Evaluating accuracy of diagnostic tests; tracking patient outcomes
Health programs and interventions	Evaluating immunization campaigns, health education and training programs
Controlling epidemics	Tracking the spread of disease; estimating mortality rates; identifying at-risk populations
Identifying barriers to care	Analysing surveys and health service usage to improve access

Application	Use of biostatistics
Health risk assessment and demography	Analysing trends in births, deaths and disease prevalence for planning services

Q.8 [Short] How is biostatistics used in epidemiology and in controlling epidemics?

In epidemiology it is used to investigate the distribution and determinants of diseases in populations: how they spread, their risk factors and the effect of preventive measures. During epidemics it tracks the spread of disease, estimates mortality rates and identifies at-risk populations, so that targeted strategies can contain the outbreak.

Q.9 [Short] What is the role of biostatistics in personalized medicine?

Biostatistical methods analyse complex genetic data, relating genes to diseases and exploring genetic variations. In personalized medicine, this lets treatments be tailored to an individual's genetic profile, to enhance effectiveness and reduce adverse effects.

22.2 KEY STATISTICAL MEASURES AND THEIR CALCULATION

Q.10 [Short] Name the key statistical measures and state what each tells.

Measure	What it tells	Kind
Mean	The average value	Central tendency
Median	The middle point	Central tendency
Mode	The most frequent value	Central tendency
Standard deviation	Spread of data around the mean	Variability
Range	Difference between maximum and minimum	Variability
Percentile	Value below which a given percentage of data falls	Distribution

Q.11 [Long] Define mean. How is it calculated for ungrouped and grouped data?

The **mean** (average) is a measure of central tendency that gives the central value of a dataset. In biology it summarizes large datasets with one typical value, such as average size or growth rate.

Ungrouped data: Mean = $\Sigma x / n$ (Σx = sum of all values, n = number of values)

Grouped data: Mean = $\Sigma f_i x_i / \Sigma f_i$ (x_i = class midpoint, f_i = class frequency)

Example (ungrouped): Weights of five birds: 400 g, 420 g, 470 g, 390 g and 460 g.

$\Sigma x = 400 + 420 + 470 + 390 + 460 = 2140$ g; n = 5

Mean = 2140 / 5 = 428 g

Example (grouped): Heights (cm) of a plant species in different soil types.

Soil type	Class interval (cm)	Frequency (f_i)	Midpoint (x_i)	$f_i x_i$
Sandy	0 – 10	5	5	25
Loamy	10 – 20	8	15	120
Clayey	20 – 30	7	25	175
Peaty	30 – 40	4	35	140
Total		**$\Sigma f_i = 24$**		**$\Sigma f_i x_i = 460$**

$$\text{Mean} = 460 / 24 = 19.17 \text{ cm}$$

In the textbook table on p.191, the total of the frequency column (24) is labelled "Median = 24"; it is $\Sigma f_i = 24$.

Q.12 [Long] Define median. How is it calculated?

The **median** is a measure of central tendency that represents the middle value of a dataset when the data are arranged in order. It represents the centre better when the data are not normally distributed, e.g., population size, disease prevalence or body measurements.

Steps: (1) Arrange the data in ascending or descending order. (2) Count the number of values, n. (3) If n is odd, the median is the value at position (n + 1)/2. (4) If n is even, the median is the average of the values at positions n/2 and (n/2) + 1.

Example (odd n): Heights of five plants: 50, 65, 60, 75, 80 cm. In order: 50, 60, **65**, 75, 80. Position = (5 + 1)/2 = 3rd value, so the median = **65 cm**.

Example (even n): Weights of six animals: 300, 400, 330, 315, 450, 500 kg. In order: 300, 315, **330, 400**, 450, 500. Median = (330 + 400)/2 = **365 kg**.

Q.13 [Short] The heights of six children are 110, 115, 118, 120, 122 and 125. Find the median.

n = 6 (even). The two middle values are the 3rd and 4th: 118 and 120. Median = (118 + 120)/2 = **119**.

Q.14 [Long] Define mode. How is it calculated for ungrouped and grouped data?

The **mode** is the value (or values) that appears most frequently in a dataset. In biology it shows the most common trait in a sample, e.g., the most frequent number of flowers on a plant or the most common seed size. For ungrouped data there is no formula; for grouped data:

$$\text{Mode} = l + [\ (f_m - f_1) / (2f_m - f_1 - f_2)\ ] \times h$$

where l = lower boundary of the modal class, f_m = frequency of the modal class, f_1 = frequency of the class before it, f_2 = frequency of the class after it, and h = class width.

Example (ungrouped): Heights of ten plants in order: 50, 50, 50, 50, 55, 55, 60, 60, 65, 70 cm. The value 50 appears 4 times, the most. Mode = **50 cm**.

Example (grouped): Weights of zoo animals: 10–20 (5), 20–30 (12), 30–40 (18), 40–50 (10), 50–60 (3). The modal class is 30–40 (highest frequency 18). So l = 30, f_m = 18, f_1 = 12, f_2 = 10, h = 10.

$$\text{Mode} = 30 + [\ (18 - 12) / (36 - 12 - 10)\ ] \times 10 = 30 + (6/14) \times 10 = 34.29 \text{ kg}$$

Q.15 [Short] What is a modal class?

In grouped data, the class interval with the highest frequency is the modal class. In the textbook example, the class 30–40 with frequency 18 is the modal class.

Q.16 [Long] Define standard deviation. Explain its calculation with an example.

The **standard deviation** measures how dispersed the data are in relation to the mean. A small standard deviation means the data are clustered tightly around the mean; a large one means they are more spread out. In biology it shows the variability in measurements such as growth rates of plants or enzyme concentrations.

$$\sigma = \sqrt{[\ \Sigma(x_i - \mu)^2 / N\ ]}$$

where σ = standard deviation, x_i = each data point, μ = mean and N = total number of data points.

Steps: (1) Find the mean. (2) Subtract the mean from each data point. (3) Square each deviation. (4) Find the average of the squared deviations (variance). (5) Take its square root.

Example: Heights of 9 plants: 56, 65, 74, 75, 76, 77, 80, 81, 91 cm. Mean μ = 675/9 = 75 cm.

x_i	56	65	74	75	76	77	80	81	91
$x_i - \mu$	–19	–10	–1	0	1	2	5	6	16
$(x_i - \mu)^2$	361	100	1	0	1	4	25	36	256

$$\Sigma(x_i - \mu)^2 = 784;\ \text{Variance} = 784 / 9 = 87.11$$

$$\sigma = \sqrt{87.11} = 9.33 \text{ cm}$$

Q.17 [Short] What do a low and a high standard deviation indicate?

A low standard deviation (close to zero) indicates that the data points are clustered tightly around the mean; the curve is narrow. A high standard deviation indicates that the data points are spread further away from the mean; the curve is more spread out.

Q.18 [Short] What is variance?

Variance is the average of the squared deviations from the mean, $\Sigma(x_i - \mu)^2/N$. The standard deviation is the square root of the variance. In the textbook example, the variance is 87.11 and the standard deviation is 9.33.

Q.19 [Short] Define range. How is it calculated?

The **range** is the simplest measure of the spread of data. It is the difference between the maximum and minimum values. A larger range means the data points are more dispersed; a smaller range means they are closely clustered.

$$\text{Range} = X_{max} - X_{min}$$

Example: Ages of a group: 25, 20, 22, 30, 42, 45, 50. In order: 20, 22, 25, 30, 42, 45, 50. Greatest value = 50, smallest value = 20. Range = 50 – 20 = **30**.

Step 4 on p.196 of the textbook reads "Subtract the maximum value from the minimum value"; the working (50 – 20 = 30) correctly subtracts the minimum from the maximum.

Q.20 [Long] Define percentile. Explain its calculation with an example.

A **percentile** is the value in a dataset below which a given percentage of observations falls. The 25th percentile is the first quartile (Q1), the 50th percentile is the median, and the 75th percentile is the third quartile (Q3). Percentiles help to understand the distribution of biological data, whether skewed or normal.

$$R = (P / 100) \times (n + 1)$$ (R = rank, P = desired percentile, n = number of observations)

If R is an integer, the value at the Rth position of the ordered data is the percentile. If R is not an integer, interpolate between the two nearest ranks.

Example: Find the 40th percentile of the heights 160, 162, 168, 165, 170, 175, 172, 178, 180, 182 cm. In order: 160, 162, 165, 168, 170, 172, 175, 178, 180, 182; n = 10.

$$R = (40/100) \times (10 + 1) = 4.4$$ (between the 4th value 168 and the 5th value 170)

$$40\text{th percentile} = 168 + 0.4 \times (170 - 168) = 168.8 \text{ cm}$$

So 40% of the observations fall below or equal to 168.8 cm.

Q.21 [Short] What is the relationship between percentiles and quartiles?

Quartiles are particular percentiles: the 25th percentile is the first quartile (Q1), the 50th percentile is the median, and the 75th percentile is the third quartile (Q3).

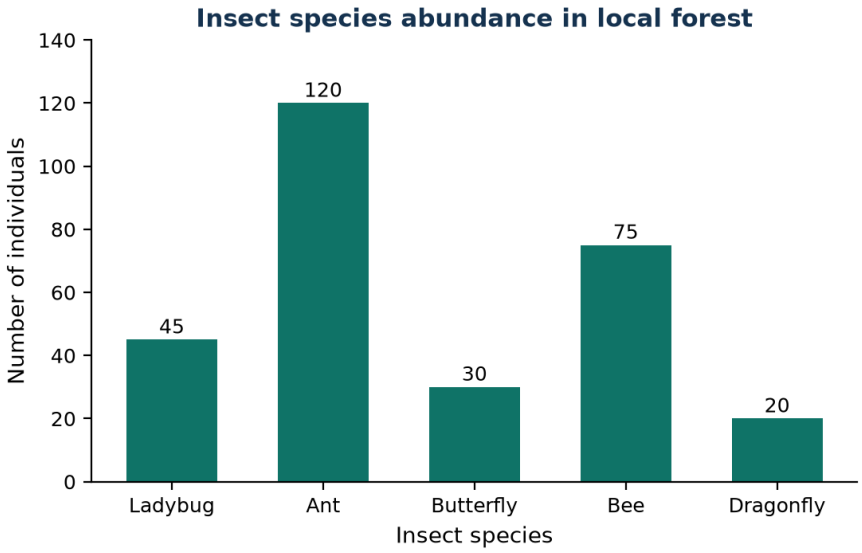
22.3 CHART

Q.22 [Long] What is a bar chart? Describe the steps to create it.

A **bar chart** (bar graph) displays categorical data with rectangular bars; the length or height of each bar is proportional to the value it represents. It is used to compare quantities across distinct categories, e.g., counts of different species or results of experimental groups. Categories are on the x-axis and values on the y-axis; bars may be vertical or horizontal.

1. **Purpose and data:** identify the categorical variable (insect species) and the numerical variable (number counted).

2. **Determine the axes:** x-axis = insect species; y-axis = number of individuals.
3. **Scale the numerical axis:** the greatest value is 120, so use 0 to 140 with an increment of 20.
4. **Draw the bars:** ladybug 45, ant 120, butterfly 30, bee 75, dragonfly 20; bars of equal width, evenly spaced.
5. **Add labels and title:** axis labels and the chart title.



Bar chart: insect species counted in a local forest (textbook data)

The textbook's own figure for this example plots the percentages (16%, 41%, 10%, 26%, 7%), while its steps describe plotting the counts. The chart above follows the steps.

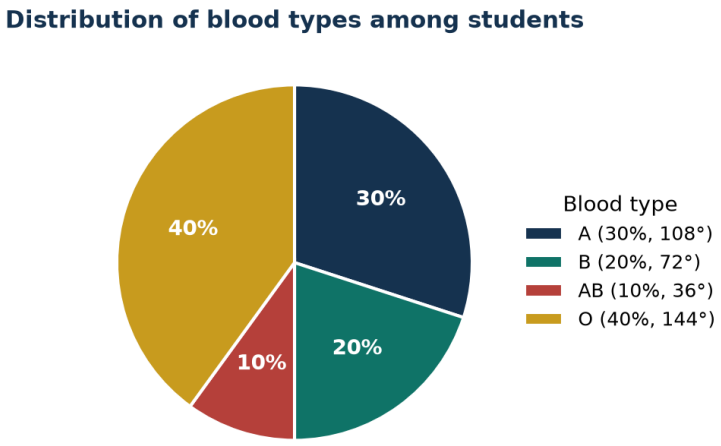
Q.23 [Short] What is a legend?

A legend is a key or guide on a chart, graph or map that explains the meaning of the symbols, colours, patterns or labels used. For example, in a bar chart with a different colour for each category, the legend identifies what each colour corresponds to.

Q.24 [Long] What is a pie chart? Describe the steps to create it.

A **pie chart** is a circular graph divided into sectors (slices), each representing a proportion of the whole dataset. The whole circle is 100% (360°), and the angle of each sector is proportional to its category's share. It is ideal for showing categories as portions of a whole, e.g., types of cells, species distribution or survey responses.

1. **Organize the data** in a table with the frequency of each category.
2. **Add up** all frequencies: 30 + 20 + 10 + 40 = 100 students.
3. **Calculate the sector angles:** sector angle = (category value / total) × 360°. A = 108°, B = 72°, AB = 36°, O = 144°.
4. **Draw** a circle; from a reference radius, mark each angle in sequence with a protractor; use different colours.
5. **Label** each sector, add a legend if needed and a descriptive title.



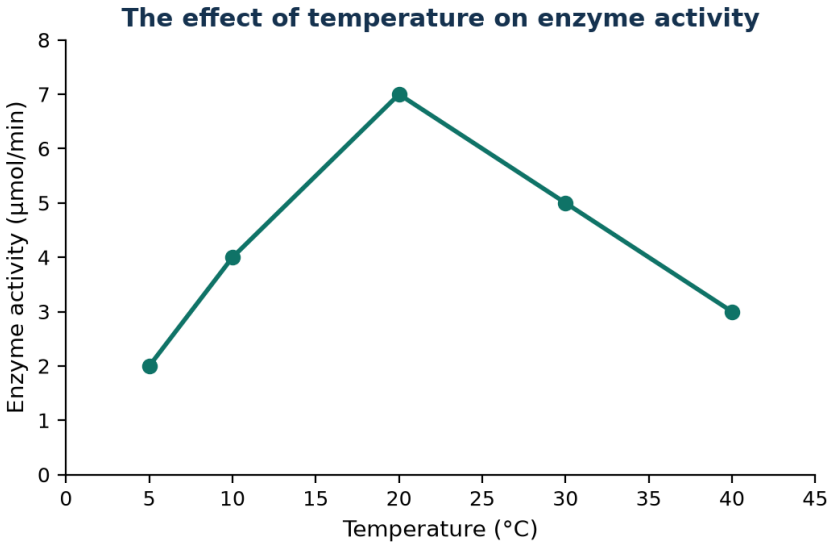
Pie chart: blood types of 100 students (textbook data)

On p.199 the textbook writes the formula as "Category value / total value × 100" but then multiplies by 360° (and prints "108° %"). For the sector angle, multiply by 360°; for the percentage, multiply by 100.

Q.25 [Long] What is an x-axis data figure? Describe the steps to create it.

The **x-axis** is the horizontal line of a graph, chart or plot. It usually represents the **independent variable** and gives the baseline against which the other variable (on the y-axis) is measured. In biology it often shows time intervals, different species or experimental conditions, and helps to show trends.

1. **Identify variables:** independent (x-axis) = temperature (°C); dependent (y-axis) = rate of enzyme activity.
2. **Determine scale and range:** 5, 10, 20, 30 and 40 °C.
3. **Label the axes:** Temperature (°C) and Enzyme activity (µmol/min).
4. **Plot the data points:** (5, 2), (10, 4), (20, 7), (30, 5), (40, 3).
5. **Adjust formatting:** evenly placed tick marks; join the points with a line to show the trend.



X-axis data figure (line graph) of the textbook enzyme data

Q.26 [Long] What are error bars? Explain their types and how they are sketched.

Error bars are graphical elements placed on bar charts (and other plots) to show the variability or uncertainty in data. They show how much the individual data points deviate from the central value (e.g., the mean) represented by the bar.

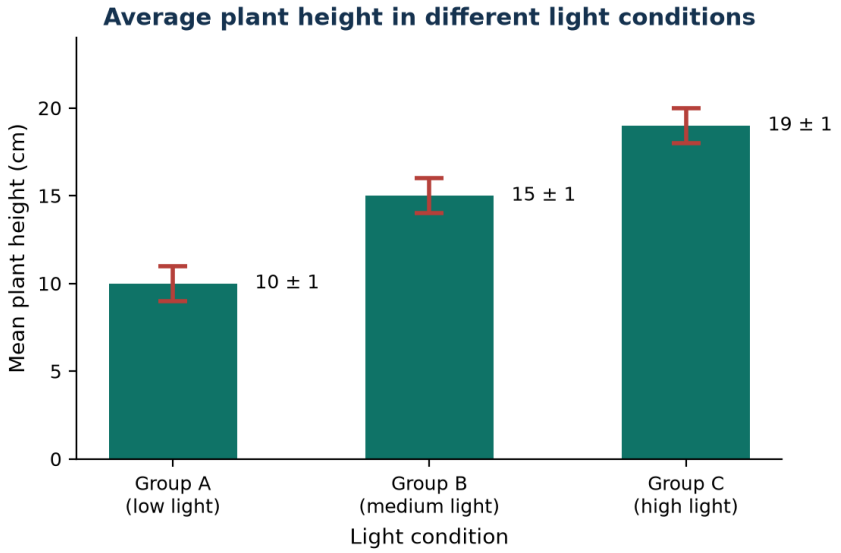
Range error bars use the smallest and greatest data points of each group; the bar extends from the minimum to the maximum value and shows the full spread of the data.

Standard deviation (SD) error bars show the average distance of the data points from the mean; they extend above and below the mean by one or more standard deviations (± SD).

Example: Heights (cm) of bean plants after four weeks in three light conditions.

Group	Plant 1	Plant 2	Plant 3	Mean	Range (max – min)	Range error (±)	SD (±)
A (low light)	10	11	9	10	2	1	1
B (medium light)	14	16	15	15	2	1	1
C (high light)	18	20	19	19	2	1	1

1. **Calculate the mean** of each group; draw the bars at heights 10, 15 and 19.
2. **Calculate the spread:** SD ≈ 1 for each group; range: A 9–11, B 14–16, C 18–20.
3. **Draw the error bars:** SD bars from mean – 1 to mean + 1 (A 9–11, B 14–16, C 18–20); range bars from minimum to maximum. Add a horizontal cap at each end.



Bar chart with error bars (here the SD bars and the range bars are equal)

Q.27 [Short] Which type of chart is appropriate for which kind of data?

Data or experiment	Appropriate figure
Quantities compared across distinct categories (e.g., counts of species)	Bar chart
Parts of one whole (e.g., blood types in a class)	Pie chart
A dependent variable measured against an independent variable (e.g., enzyme activity at different temperatures)	X-axis data figure (line graph)
Means of repeated measurements, with their variability	Bar chart with error bars

Q.28 [Short] What must a proper chart contain?

A descriptive title, labelled axes with their units, a suitable scale with evenly placed tick marks, and a legend where different colours or symbols are used.

EXPERIMENTAL DESIGN

Q.29 [Long] What is experimental design? Explain the variables and groups of an experiment.

Experimental design is the careful planning of an experiment to test an idea. Scientists decide the subjects and control the conditions so that the results are fair. Then they change one thing and see how it affects another thing, to find out whether the hypothesis is right or wrong.

Term	Meaning
Independent variable	The factor that the experimenter changes or manipulates
Dependent variable	What is measured or observed; it changes in response to the independent variable
Control variables (constants)	All other factors, kept the same throughout to ensure a fair test
Control group	The group that does not receive the treatment; it is the baseline for comparison
Experimental group	The group that is exposed to the treatment or change being tested

Q.30 [Short] Why is a control group necessary?

The control group does not receive the treatment. It serves as a baseline with which the results of the experimental group are compared. This shows whether the independent variable actually causes any effect.

Q.31 [Short] Why must control variables be kept the same?

To ensure a fair test. If other factors also changed, we could not tell whether the change in the dependent variable was caused by the independent variable or by those other factors.

Q.32 [Long] Describe the steps of experimental design.

1. State the question of the experiment.
2. Formulate a hypothesis based on existing knowledge.
3. Identify the variables: what you will change (independent), measure (dependent) and keep constant (control).
4. Set up groups: control group and experimental group.
5. Plan the procedure: how subjects are selected and assigned, how each group is treated, the duration and frequency of measurement, and how data are collected.
6. Analyse with appropriate methods (averaging, graphing or statistical tests) to compare the groups.
7. Draw conclusions; accept or reject the hypothesis on the basis of the results.

Q.33 [Long] Design an experiment to test the effect of a high-fat diet on weight gain in albino mice.

Step	Effect of high-fat diet on weight gain in albino mice
Question	Does a high-fat diet cause greater weight gain in mice than a normal diet?
Hypothesis	Mice given a high-fat diet will gain more weight than those on a normal diet.
Variables	Independent: type of diet. Dependent: weight gain. Control: age, strain, sex, cage, light, temperature, feeding schedule.
Groups	Control group: mice on normal diet. Experimental group: mice on high-fat diet.
Procedure	Select matched mice; assign them randomly to the two groups; house them in identical conditions; give the diets for a set period (e.g., 8 weeks); record weights weekly.
Analysis	Compare the average weight gain of the groups with statistical tests or graphs.
Conclusion	If the high-fat group shows a statistically significant greater gain, the hypothesis is supported; otherwise it is rejected.

In the textbook, the heading “Experimental Design” carries the number 22.3, which is already used for “Chart”.